

Strict I/O Stdout Bucket Classification

This separates cases where the original I/O oracle has no meaningful signal because both scripts print empty stdout from cases with non-empty stdout.

field	value
matched candidates	234
empty equal stdout	230
non-empty equal stdout	0
stdout differs	4

0.1. Main Findings

- Empty stdout is the whole false-positive story in this L40S clean rerun: all 156 stdout false positives have `candidate_stdout == reference_stdout == ""`.
- There are no cases where stdout is non-empty, equal, and the generated implementation is still incorrect.
- There are 4 non-empty generated stdout cases, but they do not pass the original stdout oracle because the reference stdout is empty.

0.2. empty_equal_stdout

field	value
total	230
repo stdout equal	230
stdout false positives	156
classification counts	{'not_runnable': 74, 'strict_incorrect': 156}

- G / fused_recurrent_retention: not_runnable x 4
- G / fused_recurrent_retention: strict_incorrect x 11
- G / l2_norm_bwd: strict_incorrect x 2
- G / triton_mul2: not_runnable x 2
- G / triton_mul2: strict_incorrect x 20
- T / broadcast_tensors: not_runnable x 6
- T / broadcast_tensors: strict_incorrect x 12
- T / det: not_runnable x 1
- T / det: strict_incorrect x 10
- T / dropout_sigmoid_linear: not_runnable x 3
- T / dropout_sigmoid_linear: strict_incorrect x 14
- T / elu_linear: not_runnable x 4
- T / elu_linear: strict_incorrect x 7
- T / fused_cross_entropy_log_softmax: not_runnable x 6
- T / fused_cross_entropy_log_softmax: strict_incorrect x 9
- T / fused_qr_solve: not_runnable x 3
- T / fused_qr_solve: strict_incorrect x 12
- T / fused_transformer_block: not_runnable x 7
- T / fused_transformer_block: strict_incorrect x 11
- T / invert_matrix_lu: not_runnable x 11
- T / invert_matrix_lu: strict_incorrect x 7
- T / lu.py: not_runnable x 8
- T / lu.py: strict_incorrect x 10
- T / sigmoid_adaptive_avg_pool2d: not_runnable x 8
- T / sigmoid_adaptive_avg_pool2d: strict_incorrect x 9
- T / sum_std: not_runnable x 10
- T / sum_std: strict_incorrect x 6
- T / tensordot_rsqr: not_runnable x 1
- T / tensordot_rsqr: strict_incorrect x 16

0.3. stdout_diff

field	value
total	4
repo stdout equal	0
stdout false positives	0
classification counts	{'strict_incorrect': 4}

- G / triton_mul2: strict_incorrect x 3
- T / elu_linear: strict_incorrect x 1

0.4. Non-Empty Stdout Records

0.4.1. G / triton_mul2

field	value
candidate	Bench_G_general_purpose/Qwen2.5-72B-Instruct_simp/candidate_0148.py
classification	strict_incorrect
repo stdout equal	False
candidate stdout	"Original tensor: tensor([1., 2., 3., 4., 5.])\nDoubled tensor (new tensor): tensor([2., 4., 6., 8., 10.], device='cuda:0')\nDoubled tensor (in place): tensor([1., 2., 3., 4., 5.])\n"
reference stdout	''

0.4.2. G / triton_mul2

field	value
candidate	Bench_G_general_purpose/Qwen2.5-72B-Instruct_simp_rag/candidate_0149.py
classification	strict_incorrect
repo stdout equal	False
candidate stdout	"Original tensor: tensor([1, 2, 3, 4, 5], device='cuda:0')\nResult of triton_mul2: tensor([2, 4, 6, 8, 10], device='cuda:0')\nResult of triton_mul2_inplace: tensor([2, 4, 6, 8, 10], device='cuda:0')\n"
reference stdout	''

0.4.3. G / triton_mul2

field	value
candidate	Bench_G_general_purpose/output_gpt-4o_simp/candidate_0148.py
classification	strict_incorrect
repo stdout equal	False
candidate stdout	"Output (new tensor): tensor([2., 4., 6., 8.], device='cuda:0')\nOutput (in place): tensor([2., 4., 6., 8.], device='cuda:0')\n"
reference stdout	''

0.4.4. T / elu_linear

field	value
candidate	Bench_T_general_purpose/output_DeepSeek-R1_rag/def_0146.py
classification	strict_incorrect
repo stdout equal	False
candidate stdout	'Max difference: 80.40392303466797\n'
reference stdout	''